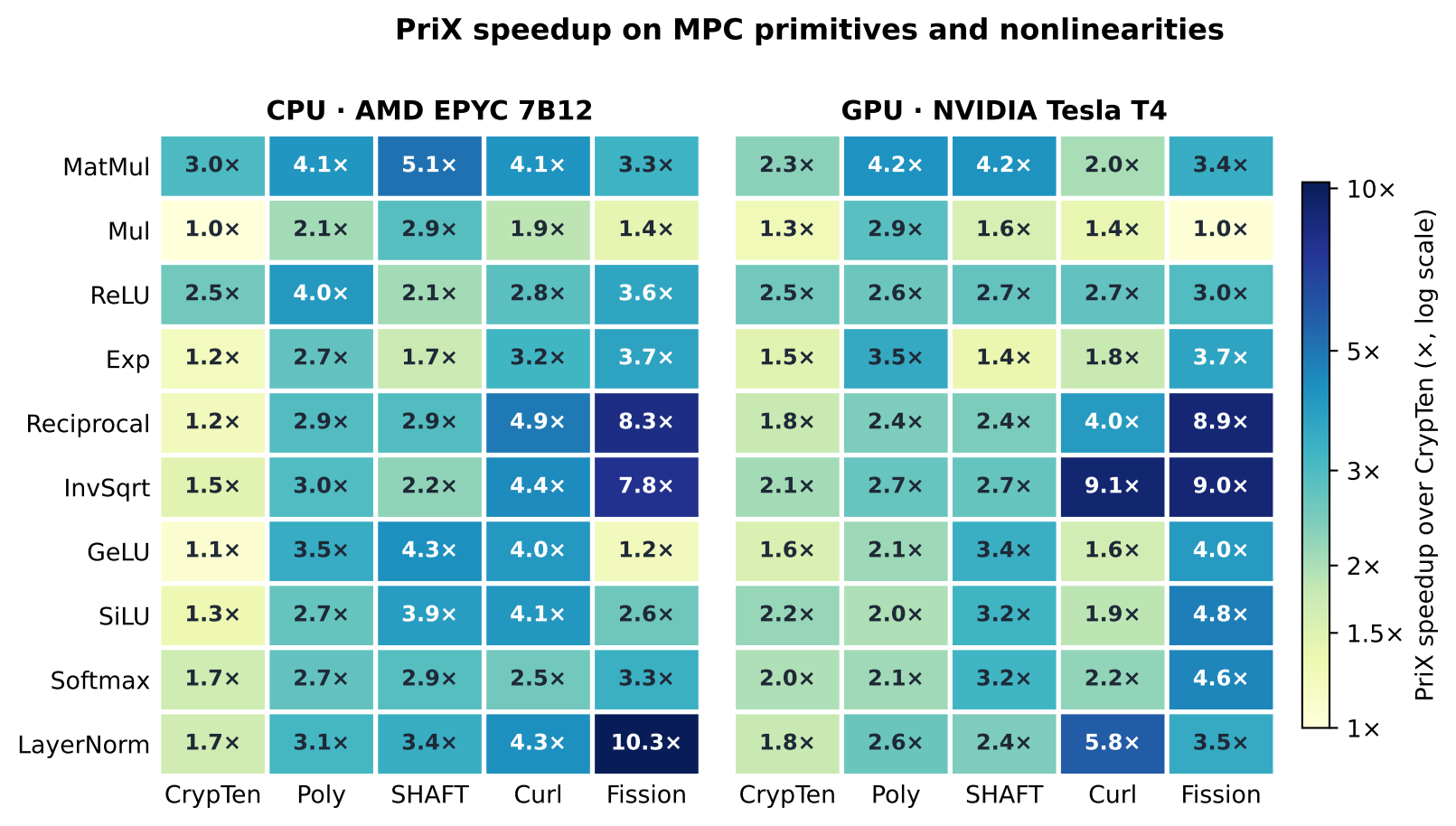
PriX speedup on MPC primitives and nonlinearities
CPU · AMD EPYC 7B12
GPU · NVIDIA Tesla T4
MatMul 3.0× 4.1× 5.1× 4.1× 3.3× 2.3× 4.2× 4.2× 2.0× 3.4×
Mul 1.0× 2.1× 2.9× 1.9× 1.4× 1.3× 2.9× 1.6× 1.4× 1.0×
ReLU 2.5× 4.0× 2.1× 2.8× 3.6× 2.5× 2.6× 2.7× 2.7× 3.0×
Exp 1.2× 2.7× 1.7× 3.2× 3.7× 1.5× 3.5× 1.4× 1.8× 3.7×
Reciprocal 1.2× 2.9× 2.9× 4.9× 8.3× 1.8× 2.4× 2.4× 4.0× 8.9×
InvSqrt 1.5× 3.0× 2.2× 4.4× 7.8× 2.1× 2.7× 2.7× 9.1× 9.0×
GeLU 1.1× 3.5× 4.3× 4.0× 1.2× 1.6× 2.1× 3.4× 1.6× 4.0×
SiLU 1.3× 2.7× 3.9× 4.1× 2.6× 2.2× 2.0× 3.2× 1.9× 4.8×
Softmax 1.7× 2.7× 2.9× 2.5× 3.3× 2.0× 2.1× 3.2× 2.2× 4.6×
LayerNorm 1.7× 3.1× 3.4× 4.3× 10.3× 1.8× 2.6× 2.4× 5.8× 3.5×
CrypTen Poly SHAFT Curl Fission CrypTen Poly SHAFT Curl Fission
10×
5×
3×
2×
1.5×
1×
PriX speedup over CrypTen (×, log scale)